Integeral Literal contd...

An integral literal we can represent in four different forms:
  a) Decimal Literal (Base is 10)
  b) Octal Literal (Base is 8)
  c) Hexadecimal Literal (Base is 16 [0-9 and a - f OR A - F])
  d) Binary Literal (Base 2) [Available from JDK 1.7v]

Decimal Literal:

By default, our numeric literals are decimal literal. Here base is 10 so, It accepts 10 digits i.e. from 0-9.

Example:
int x = 20;
int y = 123;
int z = 234;

Octal Literal:

If any Integral literal starts with 0 (Zero) then it will become octal literal. Here base is 8 so it will accept 8 digits i.e. 0 to 7.

Example:
int a = 018; //Invalid because it contains digit 8 which is out of range
int b = 0777; //Valid
int c = 0123; //Valid

Hexadecimal Literal:

If any integral literal starts with 0X or 0x (Zero capital X Or 0 small x) then it is hexadecimal literal. Here base is 16 so it will accept 16 digits i.e. 0 to 9 and A to F OR [a to f]

Example:

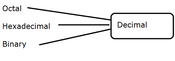
int x = 0X12; //Valid
int y = 0xadd; //Valid
int z = 0XFACE; //valid
int a = 0Xage; //Invalid because character 'g' out of range

Binary Literal:

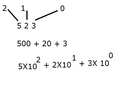
If an Integral literal starts with 0B (Zero capital B) or 0b (Zero small b) then it will become Binary literal. Binary literal is available from JDK 1.7v.
Here base is 2 so it will accept 2 digits i.e. 0 and 1 only.

Example:
---
int x = 0B101; //valid
int y = 0b111; //Valid
int z = 0B112; //Invalid [2 is out of range]

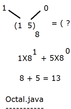
The default type is **decimal literal** so to generate the output for any different literal JVM converts into decimal literal.



//Programs :



Octal Literal program :
---
015 -> Here '0' describes that it is an octal literal.



8 + 5 = 13

Octal.java

```
void main()
{
    int x = 015;
    IO.println(x);
}
```

Working with hexadecimal :

0Xadd -> Here 0X describes that It is hexadecimal literal.

0 - 9
A = 10
B = 11
C = 12
D = 13
E = 14
F = 15

10X 256 + 13 X 16 + 13X1

2560 + 208 + 13

2781

Hexa.java

```
void main()
{
    int x = 0Xadd;
    IO.println(x);

    int y = 0xFACE;
    IO.println(y);
}
```

Working with Binary Literal :

0B101 -> 0B describes that It is binary literal

4 + 0 + 1

5

Binary.java

```
void main()
{
    int x = 0B101;
    IO.println(x);
}
```

---

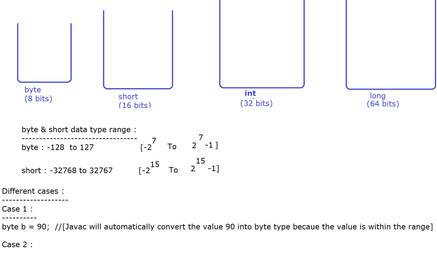
* By default, Every integral literal is of type int only.

Integeral Literal Memory Representation :
---
byte b = 5; 00000101 (8 bits)

short s = 5; 00000000 00000101 (16 bits)

int x = 5; 00000000 00000000 00000000 00000101 (32 bits)

long y = 5; 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000101 (64 bits)



byte & short data type range :
---
byte : -128 to 127 [-2^7 To 2^7 -1]

short : -32768 to 32767 [-2^15 To 2^15 -1]

Different cases :

Case 1 :
---
byte b = 90; //[Javac will automatically convert the value 90 into byte type becaue the value is within the range]

Case 2 :
---
byte b = 130; //[Compilation Error because value is out the range, range is : **-128 to 127**]

Case 3:
---
short s = 32767; //[Javac will automatically convert 32767 into short type]

Case 4:
---
long s = 9812345678; //[CE, Integer number too large (Out of the range)]

Case 5 :
---
long mob = 9812345678L; //9812345678 is of type long (64 bits)

Case 6 :

```
void main()
{
  m1(1);  //error 1 is of type int only
}

void m1(byte b)
{
    IO.println("byte");
}
```

By default, every integral literal is of type **int** only. **byte and short are below than int** so we can assign integral literal (Which is by default int type) to byte and short but the values must be within the range. [for byte -128 to 127 and for short -32768 to 32767]

Actually, whenever we are assigning integral literal to byte and short data type then compiler internally converts into corresponding type.

byte b = (byte) 12; [Compiler is converting int to byte]
short s = (short) 12; [Compiler is converting int to short]

In order to represent long value explicitly we should use either L OR l (Capital L OR Small l) as a suffix to integral literal.

According to IT industry, we should use **L** because l (small l) looks like digit 1.

Test4.java

```
/* By default every integral literal is of type int only*/

void main()
{
   byte b = 128;
     IO.println(b);

     short s = 32768;
     IO.println(s);
}
```